



REVIEW ARTICLE

Global review on interactions between insect pests and other forest disturbances

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Abstract

Context Forest landscapes worldwide are shaped by abiotic drivers such as fire, windstorms, and drought, but also by biotic drivers like insect pests and pathogens. Although the effects of such drivers on forest dynamics have been studied extensively, knowledge of the interactions between insect pests and other drivers of change is still coarse and fragmented. Indeed, new invasive insect species and global change may lead to novel interactions and

produce impacts on forest ecosystems never before experienced.

Objectives We aimed to review the mechanisms underlying interactions between insect pest outbreaks and other forest disturbances, identify interactions emerging from current disturbance dynamics, and highlight the role of simulation models in exploring these interactions in a dynamic, mechanistic, and spatially explicit manner.

Methods We reviewed the state of the science regarding interactions between insect pests and other forest disturbances, collecting a set of 216 scientific articles.

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Results Most studies focused on the interaction between insect outbreaks and fire, whereas interactions between insect pests and drought, forest management or forest diseases received much less attention. Although we identified some trends in how interactions were manifested, interactions were not more commonly found at particular spatial or temporal scales. Relatively few studies used simulation models to explore interactions between disturbances and very few studies explored multiple interactions.

Conclusions Interactions between pests and other forest disturbances play critical roles in driving forest dynamics. The effects of these interactions are likely to increase in the face of continuing global change.

Keywords Simulation modeling · Outbreak · Fire · Drought · Climate change · Landscape dynamics

Introduction

Forest disturbances such as fire, drought, windstorms, landslides, disease, and insect outbreaks are key drivers of change in forested landscapes. Disturbances modulate landscape dynamics by modifying forest composition and structure, influencing forest ecosystem functioning and resource availability (White and Pickett 1985; Seidl et al. 2017), and facilitating adaptation to new environmental conditions (Thom et al. 2017). Despite the essential role that natural disturbances play in forest ecosystem functioning and composition, our understanding of their future impacts remains limited due to the complex feedbacks that exist between disturbances and global change (Dale et al. 2001; Turner 2010). For instance, ongoing changes in the global economy, land use, and climate change may lead to more extreme disturbances (such as windstorms, floods, and wildfires) with yet unknown consequences for forest landscapes (Weed et al. 2013; Duane and Brotons 2018). Such changes may also result in novel disturbance caused by invasive species introduced via international trade (Brockerhoff et al. 2006; FAO 2008; van Lierop et al. 2015).

Insect pests are one of the main agents shaping forest landscapes, affecting almost 35 million ha annually, mainly in boreal and temperate biomes (van Lierop et al. 2015). Although many native pests

and diseases are integral agents of forest ecosystems, high-intensity outbreaks can have adverse effects on tree growth and survival (van Lierop et al. 2015). The extent and subsequent economic and ecological costs of pest outbreaks have increased in recent decades as world trade has facilitated the spread of invasive species (Hulme 2009; Turner 2010).

Insect pests are highly sensitive to global change. New environmental conditions resulting from changes in temperature, precipitation, and drought can alter insect development and reproduction (Ayres and Lombardero 2000; Kingsolver et al. 2011), and hence population dynamics and outbreaks. These changes may also enable native and invasive insect pests to spread to areas currently free of such outbreaks (Cullingham et al. 2011). Changes in land use can also influence insect population distributions and local viability (Rosenberger et al. 2017). Although the impact of direct climate-pest and land-use-pest have been examined, we know less about the indirect consequences of insect pests on forest ecosystems through their interactions with other disturbances (Ayres and Lombardero 2000; Raffa et al. 2008; Hessburg et al. 2015).

A deeper understanding of forest disturbance dynamics and their interactions is required to better forecast how global change will affect forest ecosystems (Buma and Wessman 2011). Insect outbreaks have complex relationships with other disturbances, and can be affected by fire (Parker et al. 2006; Chou et al. 2010; Hicke et al. 2012; Jenkins et al. 2014), drought (Sangüesa-Barreda et al. 2015; Temperli et al. 2015), windstorms (Reyes and Kneeshaw 2008; Potterf and Bone 2017), forest management (Bauce and Fuentealba 2013; Rosenberger et al. 2017), pollution (Roth et al. 1998; Agrell et al. 2005), and other pests or diseases affecting the forest (Jones et al. 2015; Borkowski and Skrzecz 2016). Many studies have explored the interactions between insect pests and other disturbance agents (Colgan and Erbilgin 2010; Day and Pérez 2013; Gitau et al. 2013; Anderegg et al. 2015; Millar and Stephenson 2015; Kolb et al. 2016; Agne et al. 2018; Leverkus et al. 2018). However, to date, no global synthesis exists that integrates such interactions with insect pests across different agents and regions, analyzing how global change may influence these interactions in the near future (Seidl et al. 2011), or how alterations in current ecosystems may lead to novel interaction

regimes (Turner 2010). Such a synthesis is challenged by the diversity of insect feeding strategies and outbreak dynamics, the multiple ways in which one can characterize insect outbreaks (e.g., severity, frequency, extent), and the multiple spatiotemporal scales at which disturbance interactions occur (Joseph et al. 2001; Hanula et al. 2002; Meigs et al. 2015, 2016; Kelsey and Westlind 2017a).

Ecological models are essential for improving our understanding of interactions between pests and other disturbances in the face of rapid global change. Although empirical models (based on statistical relationships among drivers and a response variable; Table 1) are widely used, they can only model observed dynamics and are therefore limited in their capacity to make predictions in novel contexts. In contrast, simulation models (mechanistic models based on a combination of theoretical understanding and mathematical/empirical information; Gustafson and Keene 2014) can be designed to include the effects of uncertain, multiple interacting disturbances characterized by cumulative effects, non-linear dynamics, cross-scale interactions and, most importantly, with the potential to capture unobserved dynamics (Clark and Gelfand 2006; Ager et al. 2007; Taylor et al. 2009; Baker and Robinson 2010; James et al. 2011; Keane et al. 2015; Maroschek et al. 2015; Leite et al. 2018). Such simulation models are particularly appropriate methods to assess changes in insect species ranges, predict novel insect outbreaks, and anticipate host–insect relationships under novel environmental conditions (Taylor et al. 2009; Maino et al. 2016; Barbet-Massin et al. 2018).

Here we present a systematic review of spatiotemporal interactions between insect pest outbreaks and other forest disturbances. We aim to: (1) detail the processes that determine how insect pest outbreaks interact with other disturbances, while highlighting the main sources of variability in such interactions; (2) identify current and potential future interactions between insect pests and emerging disturbances in the face of global change; and (3) discuss the role of simulation models as a tool for studying both current and novel forest disturbance interactions.

Methods

We searched for publications that examined interactions between insect pests and other forest disturbances. We used the Web of Science, Scopus, and Google Scholar databases to identify articles published between 1990 and 2019 that contained the following words in the title, abstract, or keyword: (Insect) AND (outbreak* OR defoliat* OR infest* OR pest OR bio* disturb* OR epidemic*) AND (forest* OR tree* OR landscape* OR stand*) AND (*fire* OR *burn* OR *drought* OR *logging* OR forest management OR harvest* OR *wind* OR hurricane OR *snow* OR *storm* OR *flood* OR *slide* OR disease OR *pollution*) AND (interact*).

We defined an interaction between disturbances as a direct or indirect relationship in which one disturbance affects the likelihood, extent, severity, or impacts of another. These effects could be either synergistic (a disturbance favors the likelihood or accentuates the impact of a subsequent disturbance) or antagonistic (a disturbance hinders, reduces, or prevents the likelihood or impact of a subsequent disturbance). We did not explicitly include climate change as a disturbance, but its effect was implicit in the influence of insect pest outbreaks, drought episodes, wildfire, wind, and flooding. Papers that were out of our thematic scope (e.g., studies focusing on agricultural instead of forest pests, or that studied multiple disturbances or climate effects on insect pests but not the interactions between them) were excluded. Finally, we included those articles that considered forest management and silvicultural interventions as drivers of change in forests, but not those that analyzed pest control via forest management. A final pool of 216 papers was selected.

We applied a common analysis scheme to all the reviewed studies. For each paper, we recorded information about the study area (at continental level), insect species examined, feeding guild (*sensu* Coviella and Trumble 1999; Labandeira 2013), disturbance agents involved, occurrence or lack of interaction, order of the disturbances, type of response, methodological approach, spatial and temporal scale, ecological explanation, and inferences regarding possible future trends of the interaction when applicable (Table 1). We categorized publications as being either empirical analyses or simulation modeling studies (see Table 1; Gustafson and Keene 2014). For simulation

Table 1 Information extracted from the selected articles

Parameter	Categories
Study area	Africa; Asia; Europe; North America; Oceania; South America
Insect species	Genus or species reported
Insect feeding guild	<i>Xylem/phloem feeders</i> insects feeding on the vascular tissue system (e.g., bark beetles) <i>Foliage feeders</i> insects feeding on leaves or needles (e.g., defoliators) <i>Other</i> gall producers, saproxylic feeders, seed feeders, or root feeders
Disturbance agents	Drought; Fire; Forest management; Landslides; Other pathogen/disease; Pollution; Snowstorm; Wind; Flood
Occurrence of interaction	<i>Lack of interaction</i> one disturbance has no effect on another <i>Occurrence of interaction</i> one disturbance effects another • <i>Synergistic</i> a disturbance increases the likelihood or impact of a subsequent disturbance • <i>Antagonistic</i> a disturbance reduces the likelihood or impact of a subsequent disturbance
Order of disturbances	Insect pest later influences another disturbance; Other disturbance later influences insect pest; Insect pest and other disturbance occur at the same time
Type of response	<i>Likelihood</i> earlier disturbance affects the probability of the later one <i>Severity</i> earlier disturbance influences the severity or intensity of the later one <i>Fitness</i> earlier disturbance influences the fitness of insect pests later in terms of reproductive success and progeny survival <i>Spread</i> earlier disturbance influences later insect spread
Methodological approach	<i>Empirical model approach</i> models assessing statistical relationships, whether causal or not, between two or more variables (also called correlative or phenomenological models) <i>Simulation model approach</i> mechanistic models built on a combination of theoretical understanding and mathematical concepts (also based on empirical information and correlative relationships) to emulate a real life system. Normally, simulation models project results to simulated future conditions. They are also called mechanistic or process-based models
Temporal scale	Days/months; 1 year; 2–9 years; 10–49 years; 50–99 years; >100 years
Spatial scale	Physiological level; Tree level (individual); Stand level (microhabitat); Landscape level
Ecological explanation	Conceptual explanation based on ecological processes that cause the interaction of different disturbances (detailed in Table 2)

The right-hand column lists in detail the different categories into which we classified each study within each information field

modeling studies, we also listed the modeling approach, the explanatory variables, the data source, and the projection scenarios considered. Information about disturbance interaction impacts (including the mechanisms of community between species, the immediate and indirect consequences of the disturbance, possible regeneration after impact, etc.) was not always available in the reviewed papers or was difficult to compare. Thus, we calculated summary statistics (frequencies) regarding the above-mentioned variables as a proxy for the incidence of interactions in ecosystems. Finally, studies that found interactions between disturbances were coded with “1”, and those that failed to find such interactions with “0”. We then used logistic regression analyses with the presence of an interaction as the binary response variable and

several predictors considered to influence the disturbance interactions (i.e., disturbance agent, order of the disturbance, type of response, insect feeding guild, spatial scale, and temporal scale) as explanatory variables. Regression models were fitted using the *glm* function in R (R Core Team 2019) and significance level was considered at p -value < 0.05.

Differences in the number of publications on disturbance interaction among continents could reflect either true geographic differences in the frequency of interactions between forest pests and other disturbances, or differences in the general amount of forest research being conducted. To account for such differences, and to control for potential geographic bias, we conducted a second search in the Web of Science looking for publications on forest disturbances, in

order to then calculate the ratio (per continent) of articles addressing insect pest interactions with other forest disturbances to articles only addressing forest disturbances.

Results

Geographic coverage

Most of the 216 papers reviewed addressed North American forest systems (173, 79%; Fig. 1). Of these papers, most focused on forest insects in western Canada and the USA (133, 61% of the total), followed by Europe (39, 18%). Studies from other continents were very rare and no studies were conducted in Africa.

With respect to articles that studied forest disturbances in general (those with and without interactions among disturbances), the greatest number were from North America and Europe (9931, 43% and 7453, 32%, respectively). The proportion of forest disturbance studies that addressed interactions with pests was similarly small across all continents: North America (1.73%), Europe (0.52%), South America (0.16%), Asia (0.13%), Oceania (0.07%), and Africa (0%).

Insect species

Xylem and phloem feeders, mainly bark beetles, were the most studied forest insect pests (71%), mainly *Dendroctonus* spp. and *Ips typographus* (Fig. 2).

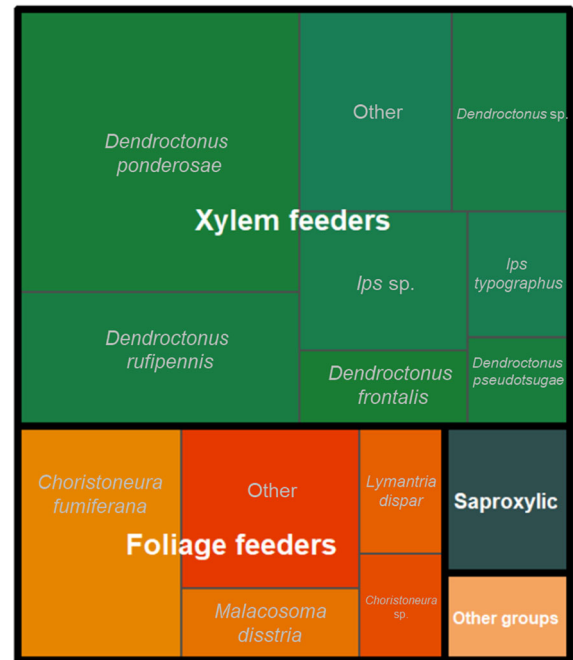


Fig. 2 Proportion of studies addressing each insect species, grouped by insect feeding guild

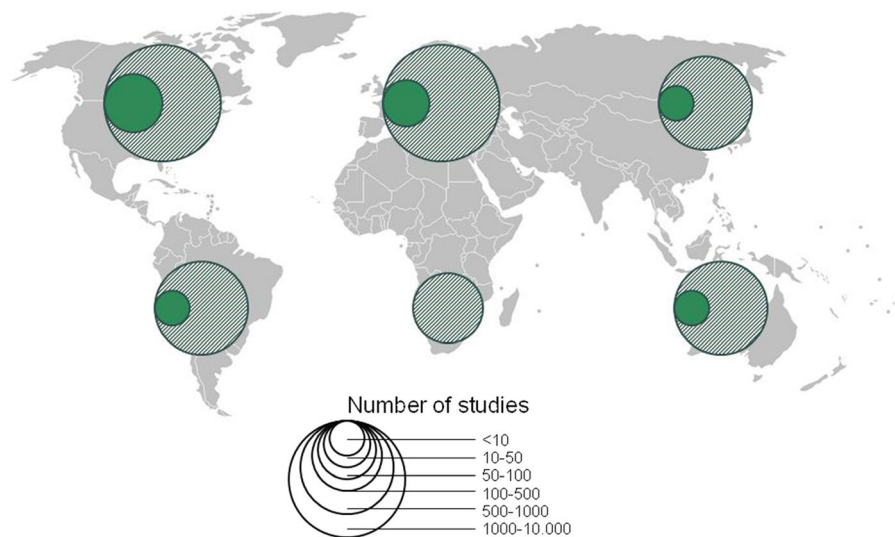


Fig. 1 Geographical distribution of number of studies at continental level. Solid filled circles indicate the distribution of the 216 papers addressing interactions between insect pests

and other forest disturbances included in the review. Striped circles indicate the distribution of papers addressing forest disturbances

Foliage feeding insects such as defoliating budworms (*Choristoneura* spp.) and gypsy moth (*Lymantria dispar*) (22%), and insects of other feeding groups (7%) were also studied. We also found that interacting disturbance agents varied with insect feeding group: fire, wind, and diseases were usually associated with xylem/phloem feeders (72%, 66%, and 64% over all studies of each disturbance, respectively), pollution was mostly studied in conjunction with foliage feeders (73%). The influence of drought, forest management, and other disturbances (landslides, floods, etc.) was considered across most feeding groups.

Disturbance interactions

Interactions between fire and forest insect pests were the most commonly studied (41%; Fig. 3). Of these fire–insect studies, 48% examined how insect-induced tree mortality affects fuel loads and consequent fire activity, 36% investigated how fire promotes insect attacks, and 16% looked at how fire and insect pests together affect forest structure and composition. The second most studied interaction was with drought (24%). Such publications investigated how water-stressed trees are more susceptible to insect attacks and how insect outbreaks can increase the

vulnerability of trees to moisture stress. Other interacting agents included forest diseases (11%), forest management (10%), pollution (6%), and wind and winter storm events (5%). Finally, 3% of studies examined other types of forest disturbances (e.g., landslides and floods). Although we did not explicitly include climate change as a disturbance in our list, it was considered in 48% of the papers examined.

Spatial and temporal scales

A similar number of studies focused on disturbance interactions at landscape scale (26%), stand scale (35%), and tree scale (30%), and only 9% of the studies focused on the physiological scale (Fig. 4). Simulation modeling studies tended to focus on landscapes (48%) and stands (44%) and much less on trees (8%; Fig. 5). Temporal scales covered a broad range, from a few days to hundreds of years, although most of the studies focused on interactions occurring between 2 and 9 years (36%) or 10 and 49 years (24%; Fig. 4). Longer temporal scales were more frequent in simulation model studies, where 10–49 years accounted for 60% of studies and 50–99 years 48% (Fig. 5).

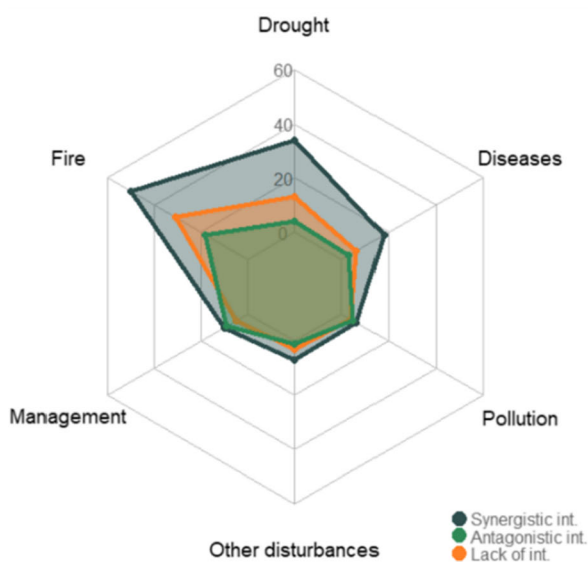
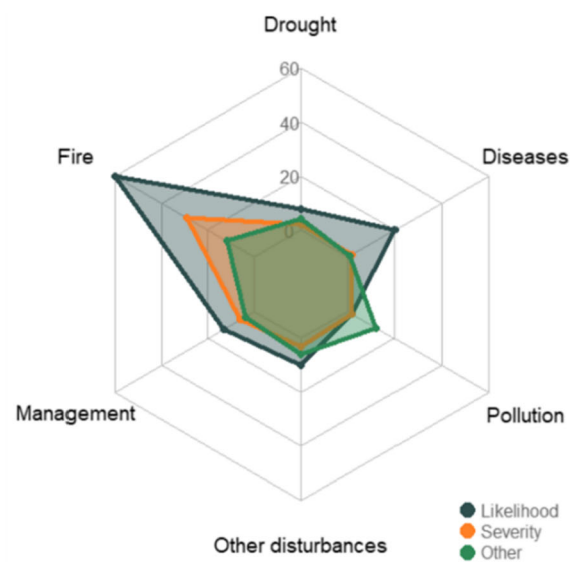


Fig. 3 Number of papers reporting synergistic interactions, antagonistic interactions, and a lack of interactions between insect pest and each type of forest disturbance (left panel).



Number of papers analyzing influences on the likelihood, severity, or another type of response between insect pests and each type of forest disturbance (right panel)

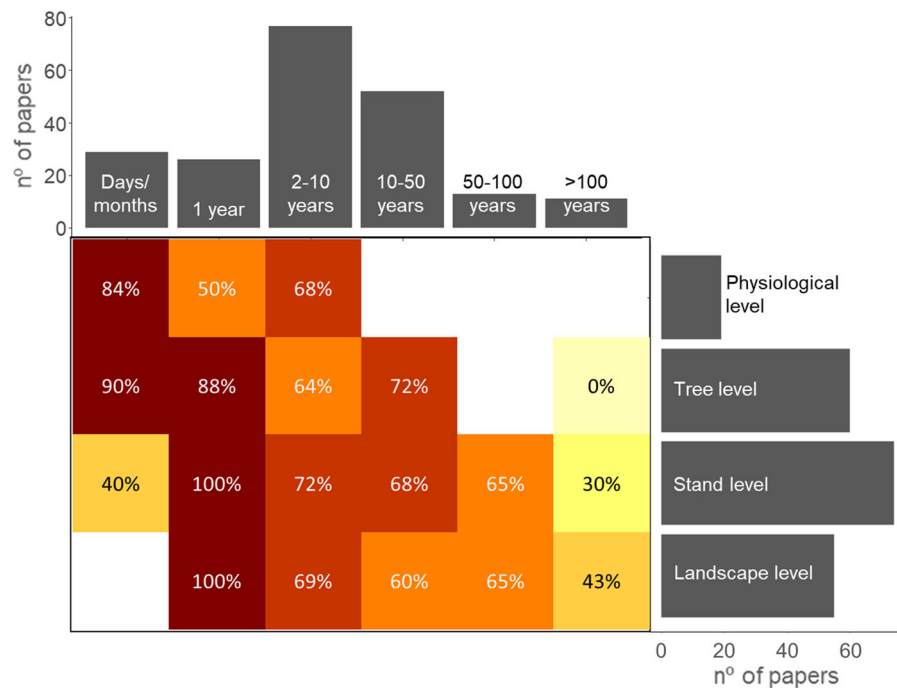


Fig. 4 Histograms of spatial and temporal scales used by reviewed studies. Heatmap indicates the percentage of interactions or lack of the interaction reported

Impact and sources of variation

Most studies reported the occurrence of interactions between disturbances (71%), considering both synergistic (54%) and antagonistic relationships (17%; Fig. 3). We found that interactions were more often identified at shorter time scales than at longer ones, while the interactions were more frequently identified at the physiological, tree, and stand scales (74%, 75%, and 71% respectively), than at landscape scale (65%; Fig. 4). However, logistic regressions showed that the presence of disturbance interactions was not significantly affected by spatial ($r^2 = 0.34$, p -value > 0.41) or temporal scales ($r^2 = 0.11$, p -value = 0.15). Similarly, relationships between temporal or spatial scales and the disturbance agent involved were not statistically significant ($r^2 = 0.18$, p -value > 0.06 and $r^2 = 0.14$, p -value > 0.13 , respectively). However, interactions between fire and insect pests were identified more often at short, than at long temporal scales ($Z = 11.15$, p -value = 0.06 and $Z = 11.03$, p -value = 0.07 at temporal scale categories 1 year and 2–9 years, respectively).

No disturbance was found to play a role in interaction occurrence more frequently than any other ($r^2 = 0.11$, p -value > 0.12 in all levels). In terms of frequencies, insect pest and forest management interactions occurred in 79% of the reviewed papers; insects and diseases, 77%; insects and drought, 74%; insects and pollution, 73%; insects and wind, 67%; insects and fire, 67%; and the two studies addressing insect pests interacting with storms and landslides also reported interactions (Fig. 3). No insect feeding guild tended to be more associated with interaction occurrence than any other ($r^2 = 0.17$, p -value > 0.91 in all levels). We also found no significant association between the occurrence of an interaction and the order of occurrence of the disturbances ($r^2 = 0.14$, p -value = 0.31). Nonetheless, we found that insects tended to follow fires in 57% of the reviewed studies, whereas fire followed insect pest outbreaks in 76% of the studies. We also found that the interaction of drought followed by xylem/phloem feeder insect pests occurred significantly more often than other combinations ($Z = 48.79$, p -value < 0.001).

Studies varied in terms of the specific effects of disturbance interactions they examined. For example,

most of the studies reviewed (59%) investigated how an initial disturbance influences the likelihood of a subsequent one and, of these studies, disturbance interactions were successfully identified in 73% of the cases (Fig. 3). Other papers (27%) examined how a disturbance affects the severity of subsequent disturbances, finding such an interaction in 70% of cases. Finally, some studies examined how multiple disturbances affect other features, like disturbance spread (10%) or competitiveness of insect pests (4%). However, logistic regressions showed that no interaction type was more associated with interaction occurrence than any other ($r^2 = 0.13$, p -value > 0.10 for any type), neither in the case of any disturbance agent in particular ($r^2 = 0.12$, p -value > 0.06).

Simulation models and interactions between disturbances

We found 23 articles (11%) that used simulation models to investigate the interaction between insect pests and other disturbances. Again, the most frequently studied interaction was between insect

outbreaks and wildfires (82%), followed by interactions with drought (35%; Fig. 5). Nine articles included three forest disturbances in a model (insect pests plus two other disturbances; Fig. 5). Almost all the models operated at either the landscape or stand scale (96%), and the most common temporal scale was 10–49 years, although some made projections over as many as 300 years. Most of the studies (63%) used previously published models (e.g., LANDIS, SORTIE, LandClim) while some built and used their own models. Many models used forest attributes (e.g., individual tree diameter, age, basal area, tree species), disturbance severity (for both insect pests and the interacting disturbance), and climatic variables as inputs. Other variables such as topography, forest spatial distribution, and soil properties were only occasionally considered. Data sources were mainly external empirical data, either from public repositories or from published studies (70%). Only in eight articles did authors compile their own data and in one case model data were simulated. With respect to scenarios, most of the studies tested models under different scenarios, either simulating different disturbance

Reference	Model	Diseases/Other pests Fire Management Other disturbances Drought Fire							<1 year 1 year 2-9 years 10-49 years 50-99 years >100 years Tree level Landscape level Forest structure Climate variables Disturbance severity Spatial variables Topography External empirical data Own empirical data Soil Simulated data Multiple climate data Multiple dist. intensity scn. Single climate scn.	
		Disturbance	Temporal scale	Spatial scale	Variables included	Data source	Scenarios			
Ager et al. 2007	FVS-PPE, Westwide Pine Beetle M.									
Chapin et al. 1997	Frame-based model									
Chew et al. 2004	SIMPPLLE									
Deroose et al. 2009	FVS-FFE, Fire and Fuels Extension									
Gustafson et al. 2010	LANDIS-II									
Hansen et al. 2015	FVS-FFE, Fire and Fuels Extension									
Hoffman et al. 2012a	WFDS									
Hoffman et al. 2012b	WFDS									
James et al. 2011	VLM									
Linn et al. 2013	HIGRAD/FIRETEC									
Loehman et al. 2017	FireBGC									
Lucash et al. 2018	LANDIS-II									
Maroschek et al. 2015	PICUS									
Nitschke et al. 2012	SORTIE									
Økland et al. 2016	Own dispersal model									
Potterf et al. 2017	IPS									
Scheller et al. 2018	LANDIS-II									
Seidl et al. 2016	FVS-FFE, Fire and Fuels Extension									
Sieg et al. 2017	HIGRAD/FIRETEC									
Simard et al. 2011	Nexus									
Sturtevant et al. 2012	LANDIS-II									
Temperli et al. 2013	LandClim									
Temperli et al. 2015	LandClim									

Fig. 5 Studies assessing insect pest interactions with other disturbances via simulation models showing the disturbances included, the temporal and spatial scales used, and the explanatory variables considered

intensities and/or the effects of different IPCC climate scenarios (83%; Moss et al. 2010). Finally, simulation-model studies reported finding interactions between disturbances at a similar frequency as the empirical studies reviewed here (68%).

Discussion

Global change is increasing the frequency and severity of disturbances and the uncertainty linked to disturbance interactions. Understanding forest ecosystem dynamics in such a context requires integrated research approaches that consider both individual disturbances and their interactive effects. In surveying the published literature regarding the reciprocal influence of forest insect pests on other disturbances, we found that interactions with fire and drought were the most frequently studied. We found that any of the factors considered (i.e., disturbance agent, type of response, insect feeding guild, or spatial and temporal scale) could be associated with the probability of an interaction being detected and reported in the scientific literature. Finally, simulation models were used in only 23 publications despite their enormous potential to help us further unravel how complex interactions among climate change, forest disturbance, and forest recovery processes contribute to forest dynamics and ecosystem resilience.

Geographic coverage and bias

Most studies of interactions between insect pests and other disturbances were conducted in North America and Europe. However, these two continents also produced most of the published papers on forest disturbances in general, with or without interactions. There are likely two main reasons for this, acting in combination: first, the reported absolute forest area affected by disturbances is larger in North America (followed by Europe, Africa, and Asia, respectively) than on other continents (Parker et al. 2006; FAO 2008; Weed et al. 2013). Second, as a result of economic and educational factors, scientific research output in general tends to be higher in North America and Europe (Smith et al. 2014; Gonzalez-Brambila et al. 2016).

The insect species represented in our survey reflect the geographic distribution of these studies, thus most

of the insect pests studied were from North America and Europe. The most common species were those that exhibit large-scale and periodic outbreaks, such as the mountain pine beetle (*Dendroctonus ponderosae*) and the eastern spruce budworm (*Choristoneura fumiferana*), as well as other Coleopteran phloem feeders such as the North American spruce beetle (*Dendroctonus rufipennis*) and the European spruce bark beetle (*I. typographus*). These species attract greater research attention as they have large impacts on forests around the globe in terms of extent and severity, and they commonly occur in regions that are also subject to fires, drought, and forest management (FAO 2008). The economic consequences of outbreaks of these species for management agencies and the forest industry are not negligible (Chang et al. 2012). However, our findings suggest that other insect pests in other regions are understudied, such as the brown Christmas beetle (*Anoplognathus chloropyrus*) and the eucalyptus weevil (*Gonipterus scutellatus*) that interact with fires in Australia (Carnegie et al. 2005; Loch and Matsuki 2010), and the dynamics of the latter (*G. scutellatus*) also being related to drought in South Africa (Graziosi et al. 2020).

Forest disturbances interacting with insect pests

Fire: insect pests

Fire is the most studied disturbance that interacts with insect pests. Although there is a well-established hypothesis that insect attacks lead to the accumulation of more fuel and a consequent amplification of wildfire susceptibility (McCullough et al. 1998), the studies we reviewed provided mixed evidence of positive, negative, and no effect of insects on wildfire (Parker et al. 2006; Jenkins et al. 2014; Meigs et al. 2015; James et al. 2017) (Fig. 2). Studies showing evidence of insect-induced fire highlight the increase in litter (Jenkins 2011; Agne et al. 2016), crown fuel accumulation (Simard et al. 2011; Hoffman et al. 2012a, b; Jolly et al. 2012; Woolley et al. 2019), vertical connectivity (Candau et al. 2018; Crotteau et al. 2018; Watt et al. 2018), flammability of needles (Jolly et al. 2012), and wind penetration due to the lack of leaves (Linn et al. 2013) as the main drivers of the interaction, leading to high risk of ignition, spread, and severity of fire (Bigler et al. 2005; Kulakowski and Veblen 2007). Other authors did not find a positive link between

Table 2 Summary of main perturbations interacting with insect pests, their interaction cause, and its ecological explanation

Perturbation	Interaction cause	Ecological explanation	References
Fire	Fuel load and connectivity	Insect-caused tree mortality or reduction of biomass moisture content alter fuel amount and connectivity, as well as the likelihood of subsequent wildfire severity	Andrus et al. (2016), Candau et al. (2018), Chapin and Starfield (1997), Chen et al. (2017), Croteau et al. (2018), DeRose and Long (2009), Donato et al. (2013), Flemin et al. (2002), Hansen et al. (2016), Hart, Schoennagel, et al. (2015), Harvey et al. (2013, 2014), Harvey, Donato, and Turner (2014), Hoffman et al. (2012a, b), Hummel and Agee (2003), James et al. (2011a, 2017), Jenkins (2011), Jolly et al. (2012), Jorgensen and Jenkins (2010), Klutsch et al. (2011), Kulakowski and Jarvis (2011), Kulakowski and Veblen (2007), Liang et al. (2016), Linn et al. (2013), Lynch and Moorcroft (2008), Makoto et al. (2012), McCarley et al. (2017), Meigs et al. (2015), (2016), Mietkiewicz and Kulakowski (2016), Mietkiewicz et al. (2018), Navarro et al. (2018), Page and Jenkins (2007a, 2007), Perrakis et al. (2014), Prichard and Kennedy (2014), Schoennagel et al. (2012), Sieg et al. (2017), Simard et al. (2011), Talucci and Krawchuk (2019), Watt et al. (2018) and Woolley et al. (2019)
	Attraction to the tree	Fire causes tree stress, leading to an increase of ethanol, monoterpenes and pheromones production that attract insects. Also, burned deadwood favors insect attacks	Beh et al. (2014), Coleman et al. (2008), Kelsey and Joseph (2003), Kelsey and Westlind (2017a), Liang et al. (2016), Lombardero et al. (2006), Veblen et al. (2006) and Westlind and Kelsey (2019)
	Tree susceptibility to insects	Fire weakens the defensive system of the surviving trees in terms of resins and other metabolites contributing to insect pest establishment	Amman and Ryan (1991), Bebi et al. (2003), Bradley and Tueller (2001), Chen-Charpentier and Leite (2014), Davis et al. (2012), Ehnström et al. (1995), Elkin and Reid (2004), Kulakowski and Jarvis (2013), Loehman et al. (2017), Lombardero and Ayres (2011), Lombardero et al. (2006), McNichol et al. (2019), Pohl et al. (2006), Powell et al. (2012), Ryan and Amman (1996), Santoro et al. (2001), Schwillk et al. (2006) and Verble and Stephan (2009)
	Forest structure and composition	A first disturbance (fire or insect pest) causes changes in forest structure (in terms of age, species, understory, etc.) and favors a second disturbance (pest or fire)	Bakaj et al. (2016), Bebi et al. (2003), Bergeron et al. (1993), Bigler et al. (2005), Boucher et al. (2018), Boulanger et al. (2013), Coleman et al. (2008), Fettig et al. (2010), Hanula et al. (2002), Johansson et al. (2007), Kerns and Westlind (2013), Kulakowski et al. (2003, 2012, 2016), Lynch et al. (2006), Lynch and Moorcroft (2008), Menges and Deyrup (2001), O'Connor et al. (2015); Perovich and Sibold (2016), Seidl et al. (2016) and Stevens-Rumann et al. (2015)
	Forest consequences	Coexistence of insect pest and fire compromises forest regeneration, that is species establishment and tree growth	Burton and Boulanger (2018), Fettig et al. (2008), Harvey et al. (2013), 2014, Harvey, Donato, and Turner 2014, Hicke et al. (2015), Kulakowski et al. (2013), Land and Rieske (2006), Liang et al. (2016), Menges and Deyrup (2001), Stevens-Rumann et al. (2015), Sturtevant et al. (2012) and Vepakomma et al. (2010)

Table 2 continued

Perturbation	Interaction cause	Ecological explanation	References
Drought	Attraction to the tree	Drought causes tree stress, leading to an increase of ethanol, monoterpenes, sugars and pheromones production that attracts insects. New climatic conditions may alter leaf palatability or deadwood accumulation, compromising insect attraction	Backhaus et al. (2014), Bolte et al. (2010), Caldeira et al. (2002), Castagneyrol et al. (2018), Haavik et al. (2015), Hale et al. (2005), Hart et al. (2014a, b), Hogg et al. (2002), Itter et al. (2019), Kelsey et al. (2014), Kelsey and Joseph (2001), Klutsch et al. (2017) and Ward et al. (2019)
	Tree susceptibility to insect	Drought weakens the defensive system of trees (e.g. resins and other metabolites, water potential) and reduced capacity of fixing nitrogen which contributes to insect pest establishment	Anderegg et al. (2015), Arango-Velez et al. (2014), Birch et al. (2019), Björkman (2000), Croise and Lieutier (1993), Dunn and Lorio (1993), Durand-Gillmann et al. (2014), Flake and Weisberg (2019), Flower et al. (2014), Gaylord et al. (2013), Jaime et al. (2019), Larsson and Björkman (1993), Lucash et al. (2018), McNulty and Boggs (2010), Moise et al. (2019), Negron et al. (2009), Pohl et al. (2006), Sangüesa-Barreda et al. (2015), Scheller et al. (2018), Suárez-Vidal et al. (2019), Temperli et al. (2015), Wermelinger et al. (2008) and Wong and Daniels (2017)
	Tree susceptibility to drought	Insect pest attacks compromise tree response to later drought disturbances	Altmann (2013), Bouzidi et al. (2019), Cailleret et al. (2017), DeRose and Long (2012), Itter et al. (2019) and Lloret and Kitzberger (2018)
	Forest structure and composition	Drought modifies forest structure and composition, directly kills trees and following insects cannot establish	Hart, Veblen, et al. (2015)
Diseases or other pests	Attraction to the tree	A first pest disturbance causes tree stress leading to an increase of ethanol, monoterpenes and pheromones production, which attracts a second pest disturbance	Aukema et al. (2006), Beh et al. (2014), Gehring et al. (2013), Grégoire et al. (2015), Kelsey and Manter (2004), Kenaley et al. (2008) and Martini et al. (2017)
	Insect vector and symbiosis	Insects and other pest have a direct or indirect symbiotic collaborative relationship that includes insects acting as vectors for other forest diseases, promoting their spread and establishment	Addison et al. (2014), Aukema et al. (2010), Ceriani-Nakamurakare et al. (2016), Firmino et al. (2017), Pinna et al. (2019), Rankin and Borden (1991), Reed et al. (2015), Shanahan et al. (2016) and Xu et al. (2018)
	Interspecific competition	Insects compete for space, resources, or protect trees from second infestations	Borkowski and Skrzecz (2016), Bylund and Tenow (1994), Jones et al. (2015), Kennedy and McCullough (2002), Kopper et al. (2004), Maňák et al. (2013), (2015), Rankin and Borden (1991), Tabacaru and Erbilgin (2015) and Tabacaru et al. (2015)
Management	Forest consequences	Landscape pattern changes due the interaction of insect pests and forest management	Fettig et al. (2008) and Mladenoff et al. (2000)
	Attraction to the tree	Management causes tree stress leading to an increase of monoterpenes production what attracts insects	Bauce and Fuentealba (2013) and Leverkus et al. (2018)
	Forest structure and composition	Forest management reduces insect pest attacks by changing tree species, structure, and density	(Zhang et al. (1993), Anhold et al. (1996), Ager et al. (2007), Johansson et al. (2007), Hayes et al. (2008), Berthiaume et al. (2009), Fettig et al. (2010), Gustafson et al. (2010), Rossi et al. (2011, 2018), Schwab et al. (2011), D'Amato et al. (2011), James et al. (2011), Temperli et al. (2014), Nowak et al. (2015), Rosenberger et al. (2017), Leite et al. (2018), Cotton-Gagnon et al. (2018) and Restaino et al. (2019))

Table 2 continued

Perturbation	Interaction cause	Ecological explanation	References
Pollution and acid rain	Tree susceptibility to insects	Host tree fitness affected by high CO ₂ , O ₃ , other pollutants, or acid rain may consequently alter later insect attacks	Agrell et al. (2005), Awmack et al. (2004), Coviella and Trumble (1999), Docherty et al. (1997), Holopainen et al. (1993), Holton et al. (2003), Kidd (1990), Kinney et al. (1997), Kozlov et al. (2017), Lindroth et al. (1993), McDonald et al. (1999), Roth et al. (1998), Roth and Lindroth (1994) and Williams et al. (1994)
	Natural enemies	Pollution demonstrate a disrupted synchrony between natural enemies and insects	Saikkonen and Neuvonen (1993)
Other disturbances	Forest structure and composition	Disturbances causes changes in forest structure and composition, compromising later insect attacks	Hanewinkel et al. (2008), Radl et al. (2017), Tabacaru et al. (2016) and Thom et al. (2013)
	Tree susceptibility to insects	Disturbances such snow, storms, wind, or floods weakens the defensive system of trees (e.g. resins and other metabolites) contributing to insect pests establishment	Angulo-Sandoval et al. (2004), Howe and Baker (2003), Hunter and Forkner (1999), Reyes and Kneeshaw (2008) and Yoneya et al. (2014)
	Insect dispersion	Wind favors insect dispersion	Havašová et al. (2017) and Potterf and Bone (2017)
	Tree susceptibility to other disturbances	Insect pest compromises tree root systems exposing them to more susceptibility to debris slides	Simard and Lajeunesse (2015)

insect outbreaks and fire, arguing that insect attacks reduce forest fuel connectivity and therefore fire activity (DeRose and Long 2009). Still other studies have suggested that the varying responses of fire ignition to insect activity is due to the different temporal scales at which researchers have looked for these interactions: when examining the effect of spruce budworm (*C. fumiferana*) outbreaks on fire ignition risk in Ontario (Canada), James et al. (2017) found that immediately following an outbreak the risk decreased but, 9 to 10 years after the outbreak, ignition risk increased. However, at broad spatiotemporal scales, studies found that other factors such as climate, forest structure, and topography had a greater influence on fire ignition risk than did the spatial legacies of past insect outbreaks (Andrus et al. 2016; Speer and Kulakowski 2017). The specific ecology of the insect species of interest (Meigs et al. 2015; Cohen et al. 2016), severity of the outbreak (Simard et al. 2011; Meigs et al. 2015, 2016), and eco-regional context (James et al. 2017) also affect fire–insect relationships.

Fires also shape the probability of insect outbreaks, directly through their effects on tree resistance and indirectly through their effects on forest structure and succession. Interactions between fire and insects affect successional trajectories by reducing regeneration

potential (Veblen et al. 2006). Alternatively, the compounded effects of fire and insects also facilitate regeneration by reducing tree competition, favoring seed dispersal (Land and Rieske 2006; Liang et al. 2016), changing forest-age structure (Arbellay et al. 2017), and replacing dominant species (Bergeron et al. 1993). A study on the spruce budworm (*C. fumiferana*) system suggests no effect of fire–insect interactions on long-term forest composition due to rapid regeneration of its primary host (Sturtevant et al. 2012).

In terms of direct effects, non-stand-replacing fires compromise tree defenses, making them more susceptible to insect attacks, due to both a reduction in bark thickness and to stress, which reduces resin production (Ryan and Amman 1996; Bradley and Tueller 2001; Santoro et al. 2001; Lombardero and Ayres 2011; Davis et al. 2012; Boulanger et al. 2013). Fire-injured trees also synthesize and accumulate ethanol, monoterpenes, and hormones that, once released to the atmosphere, act as primary attractant for some insect species such as *Dendroctonus valens*, *D. brevicornis*, *Gnathotrichus pilosus*, or *Hylurgops porosus* (Kelsey and Joseph 2003; Beh et al. 2014; Kelsey and Westlind 2017a; Westlind and Kelsey 2019), which can promote outbreaks.

Drought: insect pests

Much like fire-stress, drought influences insect-pest outbreaks because drought-stressed trees synthesize chemicals that act as insect attractants and, at the same time, may reduce their leaf- and stem-water potential limiting their resistance to insect attacks (Kelsey and Joseph 2001; Lusebrink et al. 2011; Kelsey et al. 2014; Anderegg et al. 2015; Klutsch et al. 2017). In the case of defoliators, drought stress can increase tree vulnerability as trees produce fewer palatable leaves and therefore insects need to consume more foliage to survive (Backhaus et al. 2014). Stressed trees are also limited in their ability to capture nutrients (McNulty and Boggs 2010) and produce resins, which makes them less able to defend themselves against insect attacks and therefore more vulnerable (Sanguiesa-Barreda et al. 2015; Wong and Daniels 2017).

Conversely, insect outbreaks increase tree vulnerability to water stress. When trees are attacked by insects the crown and/or roots are damaged, compromising water-regulation capacity. Thus, trees become more susceptible to later episodes of extreme drought leading to higher mortality rates (McDowell et al. 2008, 2010; Allen et al. 2010; DeRose and Long 2012; Altmann 2013; Anderegg et al. 2015; Kolb et al. 2016). Some authors have down-played the relevance of drought–insect interactions arguing that such interactions are secondary relative to other factors, such as tree fitness or the consequences of forest management (Hart et al. 2015a, b). Indeed, some studies have proposed that water stress reduces trees' vulnerability to insect attack because of the less hospitable environment insects experience in a drought-stressed tree (Hart et al. 2014a, b; Kolb et al. 2016). Yet other studies have provided evidence that defoliators can enhance tree water status by reducing canopy transpiration under drought, which is called the defoliation paradox (Bouzidi et al. 2019; Itter et al. 2019).

Diseases or other insect pests: insect pests

Insect pests also interact with other biotic disturbances, such as diseases, pathogens, and other insects. These interactions may take the form of mutualisms, such as in the case where insects act as vectors for fungal infections such as Dutch elm disease or beech bark disease (the first caused by fungus *Ophiostoma ulmi* and the second by *Neonectria faginata* and *N.*

ditissima; Rankin and Borden 1991; Aukema et al. 2010; Addison et al. 2014; Reed et al. 2015; Ceriani-Nakamurakare et al. 2016; Shanahan et al. 2016; Firmino et al. 2017; Xu et al. 2018; Pinna et al. 2019) or vectors for other bark beetle species (Croise and Lieutier 1993). As happens in the cases of fire and drought, when insects or fungi attack trees, they induce the production of volatile chemicals that attract other insects (Kelsey and Manter 2004; Aukema et al. 2006; Grégoire et al. 2015; Martini et al. 2017). However, inter-specific competition among insects or pathogens may diminish effects on host trees (Rankin and Borden 1991; Kennedy and McCullough 2002; Kopper et al. 2004; Tabacaru and Erbilgin 2015).

Forest management: insect pests

Pest control through forest management is a major topic that is out of the scope of this review. However, forest management (whether focused on pest control or not) causes alterations to forest landscapes and may interact with insect pest disturbances. Forest management affects insect outbreaks both negatively and positively (Fig. 3) and at different scales, from individual trees to landscapes (Hindmarch and Reid 2001; Ager et al. 2007; Johansson et al. 2007; Temperli et al. 2014). Thinning, prescribed burning, and commercial plantations change forest composition, landscape mosaics, and tree-age distributions. The “silvicultural hypothesis” states that forest diversity can mitigate the effects of outbreaking insects (Miller and Rusnock 1993; Jactel and Brockerhoff 2007). This hypothesis is supported by repeated observations that more diverse forest stands comprising hardwood species tend to experience less damage than homogenous coniferous stands do (Su et al. 1996; Campbell et al. 2008). Further, in a recent long-term study, Robert et al. (2018) found that the legacy of forest management strategies helped to explain the frequency, intensity, and spatial synchrony of spruce budworm (*C. fumiferana*) outbreaks. Removing dead trees, replanting new tree species, or prescribed burns interfered with outbreak development and spread (D'Amato et al. 2011; Rossi et al. 2011). Also, thinned stands were less susceptible to bark beetle species attack, likely because the plumes of pheromone the insects use for communication could not reach their targets (Thistle 2005), and because vigor and

resistance to insect attacks was improved in thinned trees (Anhold et al. 1996; Macquarrie and Cooke 2011).

Pollution: insect pests

Pollution also interacts with insect pest disturbances. Most research indicates that pollutants reduce host-tree quality as well as insect fitness, leading to a reduction in insect attacks. For instance, high ozone concentrations reduce insect fecundity and colonization rates. Increased CO₂ and heavy metal concentrations reduce growth, survival, development, and size of larvae and adults (Kopper and Lindroth 2003). Acid rain negatively affects insects directly by causing mortality, or indirectly by reducing host-plant quality (Kinney et al. 1997; Butler and Trumble 2008). However, some studies find little influence of pollution on insect performance or even a positive influence (Awmack et al. 2004). Some aphids and lepidopterans increase their growth rate and survival under high concentration of SO₂, NO₂, or O₃ (Butler and Trumble 2008). Elevated concentrations of CO₂ may play a role in insect interactions with pathogens (Roth and Lindroth 1994; Roth et al. 1998; Stiling et al. 1999) or insects' natural enemies (Percy et al. 2002), although most common responses were negative or neutral. Some studies verified that under simulated acid rain conditions, insect attacks were more severe than in the control situation (Palokangas and Neuvonen 1995; Saikkonen et al. 1995).

Other disturbances: insect pests

Other disturbances such as storms, windthrow, snow avalanches, and landslides influence insect outbreaks and their effects on forests. Together, these agents build a complex picture of direct, indirect, bidirectional, and multidirectional interactions. Such disturbances may cause changes in forest structure, compromising later insect attacks (Hanewinkel et al. 2008; Louis et al. 2014; Perovich and Sibold 2016). Also, after severe gales, storms, or landslides, the accumulation of dead wood and hence the probability of insect attacks may increase (Howe and Baker 2003; Yamazaki 2011; Simard and Lajeunesse 2015). Finally, some authors have mentioned that episodes of strong winds might facilitate the spread of insect

pests (Stadelmann et al. 2014; Havašová et al. 2017; Potterf and Bone 2017).

Climate change: insect pests

Climate change is not a forest disturbance in itself, but it has been widely recognized as a major driver of changes in insect pest regimes (Rouault et al. 2006; Bolte et al. 2010; DeRose and Long 2012; Temperli et al. 2013, 2015; Pawson et al. 2017; Rogers et al. 2017). Like many of the disturbances we have discussed, climate change influences insect pests both directly and indirectly. In terms of direct effects, increases in temperature and changes in moisture availability may increase insect survival and development rates (van Lierop et al. 2015; Malesky et al. 2018). These changes are likely to lead to shifts in geographic distributions (Friedenberg et al. 2008; Bolte et al. 2010; DeRose and Long 2012; Schwartzberg et al. 2014; Renwick et al. 2016; Marini et al. 2017; Pawson et al. 2017; Rogers et al. 2017; Jaime et al. 2019), although such positive feedback is still under discussion (Pyšek et al. 2010). In terms of indirect effects, climate change affects insect pests through direct and indirect influence on other agents (Pechony and Shindell 2010; Temperli et al. 2013; Seidl and Rammer 2017; Seidl et al. 2017).

Sources of variation in disturbance interactions

The interactions between insect pests and other forest disturbances highlighted in this review were often significant and synergistic (Fig. 3). Disturbances disrupt the structure and composition of ecosystems, create heterogeneous landscape mosaics, and change the physical environment (White and Pickett 1985; Turner 2010). In doing so, they can force ecosystem renewal, cause temporal disorganization, and alter the susceptibility of forests to new disturbances. Such synergistic interactions are especially important for the dynamics of biotic disturbances in a changing climate (Seidl et al. 2017).

In this review, we aimed to identify the main elements that favor or limit forest disturbance interactions. Different authors emphasize the relevance of the particular insect species involved in the interaction (Joseph et al. 2001; Hanula et al. 2002; Meigs et al. 2015, 2016; Kelsey and Westlind 2017a), as different species tend to have different autecologies and vary in

terms of feeding strategies, even within feeding guilds. Further, some species were studied more than others with respect to their interactions with other disturbances (e.g., the interactions between *Dendroctonus* spp. and *C. fumiferana* with fire). We therefore hypothesized that insect feeding guild (i.e., defoliators vs. xylem/phloem feeders) might be associated with particular disturbances, but the results presented here did not support this theory. Only disturbances caused by xylem/phloem feeders tended to occur more frequently following periods of drought, probably due to the higher concentration of soluble sugars in the bark of water-stressed trees resulting in improved insect larvae performance (Caldeira et al. 2002).

We hypothesized that interactions between disturbances might be more frequent at smaller spatial (at the physiological- and tree-scale) and temporal scales (1 year or less; Fig. 4). When larger spatiotemporal scales are considered, there may be more agents and processes that influence forest dynamics and disturbance and, therefore, it becomes more difficult to identify clear relationships between disturbances. Also, interactions at small spatiotemporal scales, such as positive attraction of insects at the physiological scale a few days after a fire or drought, are probably unlikely to translate into landscape-scale outbreaks (Kelsey and Joseph 2001, 2003; Kelsey et al. 2016; Kelsey and Westlind 2017a, b). However, although we identified differences in the frequency of interactions according to spatial and temporal scale, these were not significant. Thus, we concluded that scales are not correlated with the probability of identifying an interaction between two disturbances.

Studies of interactions between disturbances include a broad range of disturbance-specific features (e.g., fire disturbances are described with respect to size, severity, probability of ignition) that may affect the interaction. This is the case that Meigs (2015, 2016) and Harvey (2013, 2014) presented when they reported that insect attacks did not increase fire likelihood but altered its severity and spread. Thus, in this study we specified the type of response in disturbance interactions reported in every study (influences on the likelihood, severity, spread or insect fitness). Most of the reviewed papers that focused on insect–fire interactions examined how increased fuel loads caused by insect attacks (according to the duration of defoliation and time since the end of defoliation) facilitate subsequent fire ignition (e.g.,

Watt et al. 2020). In respect of severity, studies such as Derose and Long (2009) and Donato et al. (2013) reported a reduction in fire severity following insect attack because defoliation resulted in reduced horizontal and vertical fuel connectivity. However, our results showed that the occurrence of interactions between disturbances was not correlated with the type of response considered, either for the specific case of fire or for other disturbances. Only a small relationship between pollution and its impacts on insect fitness was found. Thus, we highlight the importance of characterizing the specific nature of disturbance interactions and the consequences on disturbed ecosystems, in addition to identifying when and where an interaction occurs.

The role of simulation models

There is increasing interest in examining the combined effects of multiple disturbances on landscape dynamics (Chew et al. 2004; Temperli et al. 2015; Seidl et al. 2016; Tabacaru et al. 2016; Lucash et al. 2018; Scheller et al. 2018). However, empirical approaches are limited in their ability to address such complex questions because past disturbances may provide an insufficient basis to understand potential future changes resulting from climate change, invasive species, and increasing human activity (Temperli et al. 2013, 2015). Simulation modeling provides one useful approach to studying future landscape changes and exploring emergent dynamics (Perera 2015).

Nonlinear, cross-scale interactions are inherent in forest landscape dynamics (e.g., interactions at tree level may leave a footprint at the landscape level; Peters et al. 2007). Empirical approaches have limited application at broad temporal and spatial scales or when processes occur across scales. Most of the modeling studies we reviewed used spatially explicit models at the scale of the landscape or stand. These models also explored longer time frames than did empirical studies (Fig. 5), of up to 300 years (James et al. 2011a; Hoffman et al. 2012a, b; Sturtevant et al. 2012; Loehman et al. 2017). The ability to explore long-term dynamics is essential in the case of insect pest disturbances because of the long term spatial legacies they can create and, in some cases, the cyclical population dynamics they generate (Fig. 5; Robert et al. 2020).

Future environmental change is uncertain, as are the dynamics of stochastic disturbances. Simulation models allow for the explicit integration of this uncertainty through scenario testing. Model-mediated exploration of such uncertain parameter space is essential to improve understanding of how different sources of uncertainty might impact the target system. Most of the articles we examined here engaged in some sort of scenario testing. (Fig. 5). Scenarios were mainly used to analyze the role of climate change in forest disturbances, usually using the IPCC climate projections (Moss et al. 2010). Other studies used scenarios to test different disturbance impacts or the response of the affected habitat (Fig. 5). Long-term projections based on a range of plausible scenarios are also valuable in facilitating policy development (Økland et al. 2016; Morán-Ordóñez et al. 2018).

While a modeling approach has benefits relative to relying on historical empirical data (e.g., the opportunity to observe unexpected emergent phenomena), developing spatial models of forest disturbance interactions can be challenging. A central challenge relates to the acquisition of high-quality data for parameterizing complex spatially-explicit simulation models, especially at large scales. Indeed, the lack of such information often stimulates the adoption of a modeling approach. Some authors collected their own data to calibrate the model (DeRose and Long 2009; Simard et al. 2011; Hoffman et al. 2012a, b; Nitschke et al. 2012; Linn et al. 2013; Hansen et al. 2015; Økland et al. 2016), whereas others used another approach, simulating the input data of the model based on expert knowledge (Temperli et al. 2015). Most modeling studies we reviewed here used data from public repositories or from other publications (Fig. 5). Long-term data monitoring and remote sensing data are particularly useful in landscape dynamics modeling because these sources offer continuous information in time and/or space (Chapin and Starfield 1997; Ager et al. 2007; Gustafson et al. 2010).

Other challenges to developing useful models of forest disturbance include integrating multiple relevant processes and their interactions at different spatiotemporal scales, as well as modeling them in a mechanistic way (Baker and Robinson 2010). Addressing these challenges requires explicit assumptions and simplifications, compromising realism while maintaining coherence, internal consistency, and plausible descriptions of modelled dynamics (Baker

and Robinson 2010; Morán-Ordóñez et al. 2018). Thus, when working with multiple disturbances, one is restricted to including only relevant landscape variables (one or a few in number) both as explanatory and response variables (e.g., stand volume, species; Clark and Gelfand 2006; Gustafson et al. 2010).

Of the 216 articles we reviewed, only 23 were based on simulation models. However, interest in using simulation modeling to explore disturbance interactions seems to be increasing: 83% of the model-based studies were published after 2010. Given the utility of simulation models mentioned above, both in scientific research and in the development of forest policies, we would emphasize the importance of the further development of simulation modeling capacity through research on how to better model complex processes and their multi-scale dynamics. To complement such model development, additional empirical information is required to improve model parameterization and model predictions, and it is the reciprocal interaction between empirical and simulation studies that will identify new challenges and opportunities and move our collective understanding forward.

Emerging disturbances and challenges to their study

Climate change, increasing global trade, and land-use change all have the potential to alter interactions between insect pests and other disturbances. Such drivers lead to an increase in invasive species that will compromise host ecosystems (Brockerhoff et al. 2006; Lovett et al. 2006; Ward and Masters 2007; Smith et al. 2012; Dix and Britton 2014; van Lierop et al. 2015; Choi et al. 2019). Some species considered to be invasive were introduced centuries ago and their effects on ecosystems are usually considered alongside native species. That is the case with the gypsy moth (*L. dispar*), which is of Euro Asiatic origin but was introduced to the US at the end of the nineteenth century (Kinney et al. 1997; Roth et al. 1998; McDonald et al. 1999). However, impacts of recent invasive insect pests are hard to predict because there is little empirical evidence about their performance in new environments with new competitors. One such species is the box tree moth (*Cydalima perspectalis*) native to Asia, which has been invading European forests since 2007 (Bras et al. 2019). Its defoliating effect leads to the death of box trees (*Buxus*

sempervirens), which are abundant in the understories of European forests. This may cause an increase in fuel availability and consequently affect the severity of fire risk or, conversely, it may reduce the probability of fire spread by eliminating the main understory species and reducing fuel connectivity.

Land-use change poses another challenge to understanding interactions between insects and other forest disturbances. It is broadly assumed that the replacement of primary forests with pastures and croplands causes a reduction in insect biodiversity, as seen in different ecosystems around the world (Koh 2007; Almeida et al. 2011; Meijer et al. 2011; Korasaki et al. 2013). Moreover, Barragán et al. (2011) documented a loss in beetle functional diversity (based on food relocation, body size, daily activity period, and food preferences) as a result of land-use changes in Mexico. However, the study of land-use effects on insect pests is mainly focused on agriculture (Kiritani 2007; van Lierop et al. 2015), and few studies investigate forest insects in the context of land-use change (Rosenberger et al. 2017). An example may be the pine processionary moth (*Thaumetopoea pityocampa*), whose growth is linked to habitat type (Torres-Muros et al. 2017). Given rural abandonment and pine reforestation in the natural range of this insect species (Pausas et al. 2008; Cervera et al. 2019), an increase in pine processionary moth pest risk is possible. Thus, researchers and managers should acknowledge that current dynamics and management policy effects extend beyond the short-term and local scale and should pay special attention not only to current forest pests but also to those that may have future impacts (Table 2).

Finally, it is essential to improve theoretical knowledge of outbreaking insect species (genetics, population dynamics, species ecology, and distribution) to better understand the mechanisms behind interaction dynamics (Ayres and Lombardero 2000; Loehman et al. 2017; Xu et al. 2018). Other authors have highlighted problems with obtaining high-quality data for such studies, because long-term and broad-scale monitoring is needed (Hanula et al. 2002; Aukema et al. 2010; Gustafson et al. 2010). New technologies and platforms, such as data obtained using satellites or LiDAR, as well as improved demographic and population genetic data, represent enormous resources with which we can better parameterize spatially-explicit simulation models (Chou et al. 2010; Hollaus and Vreugdenhil 2019).

Conclusions

Our review summarizes the current understanding of interactions between insect pest outbreaks and other forest disturbances and highlights the complexity of these processes. Such interactions are critical drivers of landscape dynamics in many forested systems. Reported interactions between disturbances were synergistic, antagonistic, or not detected. They were detected at different spatial (from tree to landscape level) and temporal (from days to century) scales and included various types of interaction (influences on the likelihood, severity, spread or insect fitness). However, we found no clear relation between such disturbance features and the occurrence of interactions.

The impacts of insect pest outbreaks are expected to increase with intensifying global change. The fitness and distribution of insect species are strongly influenced by climate. Furthermore, insect pest outbreaks can interact with pollution and land-use changes, and invasions by insect species are increasing as a result of global trade. Understanding the potential for altered interactions between insect pests and other disturbances as well as the emergence of novel interactions are major challenges in forest dynamics research. In this context, simulation models and spatially explicit simulations of potential future global-change scenarios may play an important role in the management of disturbance interactions and landscape dynamics because they can incorporate multiple drivers that operate at broad spatiotemporal scales and can be used to project possible future scenarios. Such predictions of future ecosystem conditions must be supported by appropriate theoretical knowledge and must be relevant to decision-making processes.

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search and data analysis was performed by CQ. The first draft of the document was written by CQ and all authors (CQ, AN, LJJ, JPMa, BL) commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability All data generated or analysed during this study are included in this published article and its Supplementary Information files.

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